

Task-Aware Reuse of Cellular Molecular Receivers: Response Prediction and Feedback-Driven Waiting

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Abstract—A living cell used as a molecular receiver remembers earlier commands, so a command sent too early can fail. We ask what information predicts a cell’s reporter response under repeated commands and whether feedback can replace a fixed guard interval. On 1,565 experimental FGF2–ERK single-cell trajectories, reporter filtering plus a calibrated mechanistic (B3) increment beat persistence on the validation and three test protocols (by 6.7–7.6% in 10-min RMSE); the increment carried this gain (4.5–9.8%; five of six held-out protocols), whereas filtering alone helped on two and input history added at most 2%. In the next-command windows of two test conditions, however, it raised the pooled 10-min error by 7.2%, driven by 25 ng/ml, where a later probe is consistent with B3 overestimating recovery after a long command. In B3 simulations requiring 90% reliability per previous command (noise s.d. 0.005), waiting on a smoothed reporter or the B3 belief finishes 5.5–7.7 min earlier than history-conditioned fixed waits (three calibration targets); with a feedback delay the rule must predict readiness, and if B3 is the true receiver the mechanistic estimate pays off at high noise and, less robustly, with 10-min sampling. Feedback gains by probing soon after readiness: a 2-min recovery lag hidden from the reporter broke the requirement for these two rules, a 4-min lag for fixed waits, whereas the measured probe corresponds to ≥ 35.5 min. A guard time after the trigger, calibrated on complete simulated outcomes, restored the requirement up to a 12-min lag, 6.2–7.7 min faster than recalibrated fixed waits (an upper bound).

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, Bayesian filtering, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living cells as receivers offers a natural interface between engineered communication systems and biological processes. Unlike a conventional receiver, a cell retains the effect of earlier chemical stimuli in receptor dynamics, intracellular signaling and feedback, so repeated commands are not independent: a receiver that has just responded may not execute the next command with the same reliability. This raises a communication-level question for repeated molecular tasks: *when can a living receiver be used again, and what information does that decision need?*

Receiver memory is familiar in MC, but it is usually placed in the channel. Diffusion leaves residual molecules that cause intersymbol interference (ISI) [1], [2], and receivers counter it with sequence detection and equalization [3], decoders that exploit memory and sampling rate [4], adaptive thresholds [5], and symbol or detection intervals that trade rate against residual ISI [6], [7]; ligand-binding reception adds its own

noise [8]. In these designs the guard interval is set in advance. For a cellular receiver the memory is a hidden biochemical state seen only through a noisy reporter, and deciding from that reporter when to send the next command is a stopping problem, related to sequential detection [9], [10] and optimal stopping [11]. History-dependent FGF2–ERK responses [12] and frequency-dependent signaling [13] show that such memory exists, feedback control of single cells is feasible [14], [15], and experimental and computational work on biological MC systems is emerging [16]–[19]. What is missing is a measurement-based account of which information predicts a cell’s response under repeated commands, including the next one, and a test of whether feedback can replace a fixed guard interval.

We combine experimental FGF2–ERK single-cell trajectories with the calibrated B3 signaling model of [12]. Nested predictors add the current reporter, causal reporter filters, lagged values, a mechanistic B3 increment and explicit input history one at a time. Receiver reuse is posed as a calibrated stopping rule on B3 with a reliability requirement for each previous command. Two links connect the parts, although the rules do not use the experimental predictors: with a feedback delay the rule must predict readiness ahead, as the forecasts do, and a measured probe that suggests B3 overestimates recovery after a long command motivates a stress test of the waiting rules under a recovery lag that the reporter does not show. Our contributions are:

- On measured cells, reporter filtering plus the B3 increment beats persistence on the validation and all three test protocols (by 6.7–7.6% in 10-min RMSE), and the increment carries this gain: it lowers the 10-min error of the filter on all four (by 4.5–9.8%) and in five of six held-out protocols, and on Test A lifts the noise-adjusted skill at 20–30 min to 21% (9–13% with filtering alone); at 10 min reporter filtering helps on the validation protocol and Test A only, and input history adds at most 1% in RMSE on the test protocols (up to 2% across held-out protocols) and destabilizes longer forecasts on Tests A and B. In the next-command windows of two test conditions the B3 increment raised the pooled 10-min error, through the 25 ng/ml condition, and a later probe pulse in those cells is consistent with B3 overestimating recovery after a long command; a posterior predictive check places the measured recovery after a 30-min pulse

at 25 and 250 ng/ml below every B3 posterior sample, whereas B3 matched it at 2.5 ng/ml and after short pulses at the doses it was fitted to.

- Under a per-history 90% reliability requirement at the reference noise (s.d. 0.005), waiting on a smoothed reporter or the B3 belief finishes 5.5–7.7 min earlier than history-conditioned fixed waits and closes 38–45% of their gap to the oracle across three pre-specified calibration targets (42–45% at 0.94, the only target at which every rule met the requirement in all splits); at this noise the smoothed reporter is as good as the B3 belief, and the raw current reporter gains nothing. This speed comes from probing soon after readiness, so it relies on the model’s recovery timing; under a recovery lag hidden from the reporter, the smoothed and B3-belief rules kept the requirement only without lag and fixed waits and the current-reporter rule up to 2 min, whereas the measured probe corresponds to a lag of at least 35.5 min, at which even the oracle succeeds for at most 31% of receivers after the long command; a guard time after the threshold crossing, calibrated with the threshold on complete simulated outcome curves, restored the requirement for lags up to 12 min at 6.2–7.7 min less completion time than recalibrated fixed waits, an upper bound for calibration from measured probes.
- A condition map over feedback delay, sampling interval and noise shows when a correct mechanistic model is worth using: readiness must be predicted rather than nowcast once the delay reaches about 20 min, and the B3-predictive rule beats a smoothed reporter with its trend by 0.9–5.4 min at high reporter noise (target 0.94; at similar success in 11 of 15 map cells at targets 0.90 and 0.94, 9 at 0.92) and, less robustly, by up to 1.1 min with sparse sampling at the reference noise, but not at low noise or with dense sampling at the reference noise; these gains, like those of Contribution 2, presuppose that the model’s recovery timing is right.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ be the commanded FGF2 concentration (ng/ml). Delivery to the receiver is represented abstractly as

$$\begin{aligned} \tilde{u}(t) &= \mathcal{T}_\theta[u](t), \quad \dot{x}_j = f_{\theta_j}(x_j, \tilde{u}), \\ z_j &= g(x_j), \quad y_{j,i} = z_j(t_i) + \nu_{j,i}, \end{aligned} \quad (1)$$

where $\tilde{u}$ is the local concentration, x_j the hidden state of cell j , z_j its noise-free normalized ERK reporter and $y_{j,i}$ the observation at frame t_i with noise $\nu_{j,i}$. Neither $\mathcal{T}_\theta$ nor $\tilde{u}$ is calibrated here. The information $\mathcal{H}_t$ holds the executed commands and the observations that have arrived by t .

Reuse task. The amplitude and duration of the next command are fixed; only its timing is decided. A previous command of $m \in \{3, 30\}$ min ends at time 0, and a 5-min probe at the same amplitude acts on the receiver at $\tau \in \{0, 2, \dots, 120\}$ min. With z_j^τ the response when the probe

TABLE I
NESTED PREDICTORS AND THEIR FEATURES.

Predictor	$u_t, y_t, \text{ slope}$	$\bar{y}_\ell$	lags	$\hat{\delta}_{t,h}$	$v_\ell, \int u$
Persist ($\hat{y}_{t+h} = y_t$)					
Current	✓				
Filter	✓	✓			
Lag	✓		✓		
Filter+B3	✓	✓		✓	
Lag+B3	✓		✓	✓	
F+B3+Input	✓	✓		✓	✓

acts at τ and r_j^{nv} the same receiver’s rise to the probe without a previous command,

$$\begin{aligned} S_j(\tau) &= \mathbf{1} \left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_j^{\text{nv}} \right\}, \\ R_{j,t} &= \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t), \end{aligned} \quad (2)$$

with window $W = 20$ min. Success is scored on the noise-free response; decisions see only noisy observations. The completion time is $T = \tau + W$ on success and $T_f = 140$ min on failure, the latest completion of any success. The design problem is

$$\min \mathbb{E}[T \mid m] \quad \text{s.t.} \quad \mathbb{P}(S = 1 \mid m) \geq q, \quad m \in \{3, 30\}, \quad (3)$$

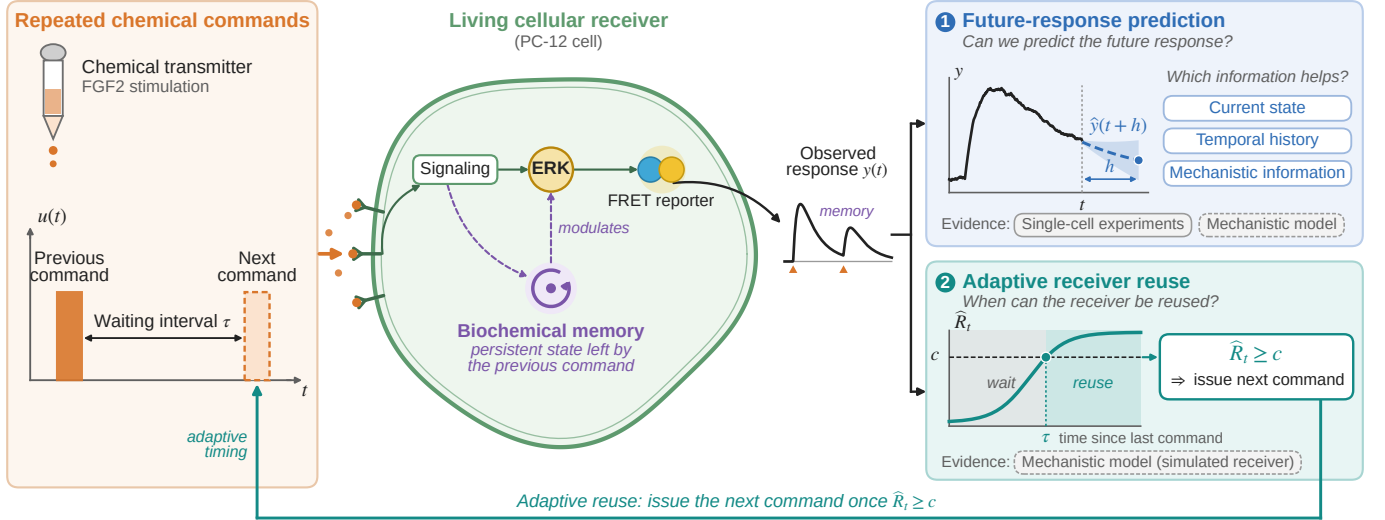
with $q = 0.9$ the required reliability, checked empirically on held-out samples rather than guaranteed; rules may be calibrated to higher targets to meet it (Sec. V-C). Requiring q for each previous command, not on average, prevents a rule from buying speed after one history with failures after the other. We use $\kappa = 0.5$, fixed before the waiting analyses after $\kappa = 0.8$ proved infeasible; a later scan over $\{0.30, 0.35, \dots, 0.70\}$ shows that it is also the largest value at which an oracle, which probes at the first decision time at which the probe would succeed, still succeeds for $> 99\%$ of simulated receivers within 120 min, pooled over both histories (99.25%, 98.5% after the 30-min command; 95.1% at 0.55, 82.0% at 0.60). If the receiver’s state reaches the controller late, or the command takes time to arrive, a probe decided at t acts at $t+d$: the rule must predict R at $t+d$ from $\mathcal{H}_t$, a forecast with horizon $h = d$.

III. PREDICTORS AND WAITING RULES

A. Nested Predictors

Predictors for horizons $h \in \{2, 10, 20, 30\}$ min start after a fixed 10-min warmup; future reporters are targets only. Table I lists the feature sets (slope: last reporter increment per minute; $\int u$: cumulative command; repository code keys P, C, O, A, M, A+M and H, in table order); every learned predictor also receives the known future command and is fitted by ridge regression with equal weight per condition. Lags are the reporter and command values of the previous 5–30 frames. The causal reporter and input filters with scales $\ell \in \{2, 10, 60\}$ min are

$$\begin{aligned} \bar{y}_{\ell,i} &= e^{-\Delta_i/\ell} \bar{y}_{\ell,i-1} + (1 - e^{-\Delta_i/\ell}) y_i, \\ v_\ell(t) &= \frac{1}{\ell} \int_0^t e^{-(t-s)/\ell} u(s) ds, \end{aligned} \quad (4)$$



Potential applications: biosensing · closed-loop biochemical control · adaptive therapeutic stimulation · synthetic biological communication

Fig. 1. System concept. Repeated FGF2 commands drive a living receiver whose biochemical memory shapes the ERK/FRET response $y(t)$ to later commands. The response supports (1) prediction of $\hat{y}(t+h)$ and (2) readiness-based reuse, where the next command is sent once $\hat{R}_t \geq c$.

with $\bar{y}_{\ell,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$. As mechanism we execute the authors' B3 model with their 1,000 posterior samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by the cell's past reporter,

$$w_s(t) \propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (5)$$

$$\hat{\delta}_{t,h} = \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. Since $\hat{\delta}_{t,h}$ already uses the command history, F+B3+Input tests only the extra value of explicit input features. Penalty, σ and lag length are chosen on validation data; normalization, lag filling and fitting details are in the repository.

B. Calibrated Waiting Rules

A waiting rule probes at the first decision time at which its readiness estimate reaches a threshold c , or at 120 min,

$$\tau = \min (\{t : \hat{R}_t \geq c\} \cup \{120\}), \quad (6)$$

a stopping time on the observations, as in sequential tests [9]. Three estimators of R_t are compared on B3: *current* uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; *smoothed* applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the *B3 belief* weights reference receivers under the known command history by all observations, as in Eq. (5), with likelihood s.d. equal to the noise s.d. floored at 0.001. The comparator is one fixed wait per previous command, a history-conditioned guard interval. References are one half of the posterior samples (calibration; each receiver left out of its own references); the other half (evaluation) never enters references, thresholds or waits. Every rule, fixed or adaptive, chooses its parameters per previous command (a wait, a threshold, or for

smoothing a scale and a threshold) to minimize that history's calibration completion subject to its calibration success $\geq$ a target, i.e. Eq. (3) on calibration samples. Twenty splits redraw the halves and the observation noise.

With delay d , the *B3-predictive* rule averages the references' success at $t+d$ under the same weights, $\hat{R}_{t+d|t} = \sum_s w_s(t) S_s(t+d)$; the *B3 nowcast* ignores the delay. The comparator without a command-history model, *smoothing plus slope*, also extrapolates: a kernel over the smoothed level and its last increment, $(\bar{y}_t, \bar{y}_t - \bar{y}_{t-\Delta})$, with the covariance of smoothed white noise, maps the current trend to success at $t+d$. Observations arrive every $\Delta \in \{2, 6, 10\}$ min and decisions are made when they arrive.

IV. DATA AND EVALUATION

All numerical files come from the paper-linked repository at a pinned commit [20], [21]; commands follow the authors' model XML, and per-cell normalization uses only frames before stimulation onset. Table II lists the 22 conditions with their roles and cell counts (1,565 cells in total, 2-min frames). Roles were frozen before any newly used response file was downloaded: Test A is a new protocol, Test B's timing was inferred from the authors' simulations, and Test C uses new concentrations recorded in the session of the inspected mixed blocks (role "Viewed"). Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and mixed blocks at 2.5 and 250 ng/ml ("fit" and "arch." in the last column of Table II), B3 is a calibrated reference, not a model blind to those protocols; it is executed from the authors' files without refitting (posterior-predictive mean within 8.5×10^{-6} of their exports). In the corrected leave-one-protocol-out (LOPO) each of six protocol groups is held out with predictor tuning restricted to the training

TABLE II
PROTOCOLS, ROLES AND CELLS.

Protocol	Role	ng/ml	Cells	B3 use
Sustained	Train	0.25, 2.5, 25, 250	248	fit 2.5, 250
3/20 pulses	Train	2.5, 25, 250	207	fit 2.5, 250
Single 5 min	Val.	0.25, 2.5, 25, 250	370	arch. 2.5, 250
Single 10 min	Test A	0.25, 2.5, 25, 250	248	–
Single 60 min	Test B	2.5, 25, 250	167	–
Mixed	Test C	0.25, 25	171	–
Mixed	Viewed	2.5, 250	154	arch. 2.5, 250

TABLE III
FORECAST ACCURACY ON MEASURED CELLS.

(a) 10-min RMSE by role

Model	Val.	A	B	C	A, 2 min
Persist	0.02353	0.02287	0.02651	0.02876	0.01302
Current	0.02503	0.02442	0.02715	0.02825	0.01342
Filter	0.02403	0.02220	0.02715	0.02827	0.01283
Lag	0.02343	0.02272	0.02565	0.02769	0.01300
Filter+B3	0.02197	0.02120	0.02448	0.02658	0.01263
Lag+B3	0.02161	0.02114	0.02474	0.02652	0.01270
F+B3+Input	0.02195	0.02098	0.02439	0.02656	0.01260
Filter < Current	4/4	4/4	2/3	1/2	4/4
Filter+B3 < Filter	4/4	3/4	2/3	2/2	3/4
Lag+B3 < Lag	4/4	4/4	3/3	1/2	4/4
F+B3+Input < Filter+B3	1/4	2/4	2/3	1/2	2/4

(b) 10-min RMSE, leave-one-protocol-out

Held-out	Filter	F+B3	F+B3+In	Lag	Lag+B3	B3 use
Sustained	0.02102	0.01969	0.01941	0.02025	0.01931	fit
3/20 pulses	0.03680	0.02886	0.02830	0.03287	0.02837	fit
Single 5 min	0.02243	0.02153	0.02139	0.02196	0.02134	arch.
Single 10 min	0.02084	0.02106	0.02104	0.02116	0.02082	none
Single 60 min	0.02701	0.02410	0.02369	0.02486	0.02422	none
Mixed, 0.25/25	0.02709	0.02623	0.02610	0.02706	0.02628	arch. ^o

protocols. Forecast rows are assigned to the phases stimulation, next command (a pulse starts in $(t, t + h]$), early washout (≤ 30 min after a pulse) and late. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}(h) = \left[\frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{1}{N_{k,h}} \sum_{i \in k} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (7)$$

over the conditions $\mathcal{K}$ of a role. Because every error contains the reporter noise, we also report a noise-adjusted skill, $1 - (\text{MSE} - \hat{\sigma}_\nu^2) / (\text{MSE}_{\text{Persist}} - \hat{\sigma}_\nu^2)$, with $\hat{\sigma}_\nu$ a robust second-difference estimate of white reporter noise per condition (0.006–0.009), and 95% intervals from resampling cells within conditions. Cells share a session and no day identifiers exist, so intervals describe cell sampling, not independent experiments. Every analysis was frozen in its own protocol before running, but all are post hoc with respect to inspected data.

V. RESULTS AND DISCUSSION

A. Which Information Predicts the Response

Table III(a) reports the RMSE of Eq. (7) at 10 min for the validation (Val.) and test roles (A–C) of Table II (last column: 2 min on Test A); bold marks the best predictor per

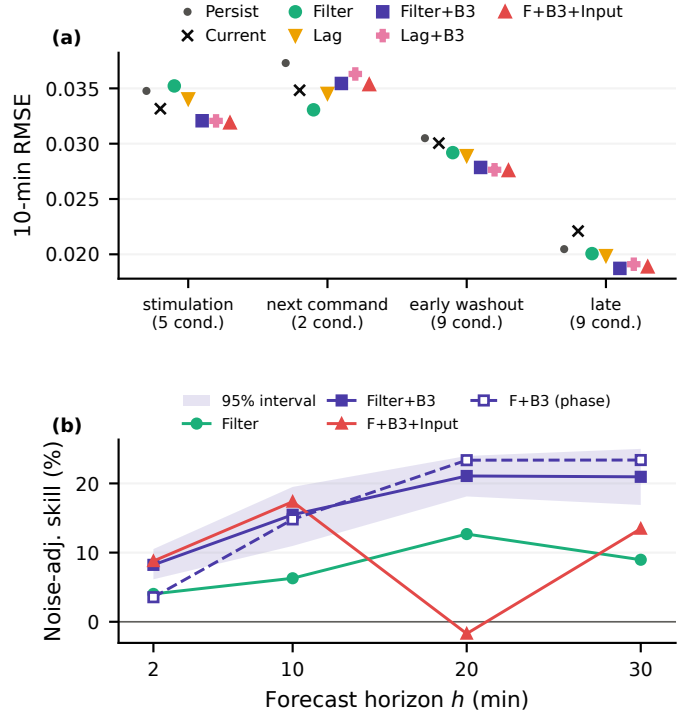


Fig. 2. (a) Condition-equal 10-min RMSE on Tests A–C by forecast phase (axis not starting at zero). (b) Noise-adjusted skill on Test A by forecast horizon; band: 95% cell-bootstrap interval of Filter+B3.

column, and the bottom rows count the conditions in which the first predictor of each pair has the lower MSE. The most robust result is that Filter+B3 beat persistence on all four roles, by 6.7–7.6%, and that the B3 increment carried this gain: it lowered the error of Filter on every role (4.5–9.8%), whereas filtering helped only on the validation protocol and Test A (4.0% and 9.1%; none on Tests B and C at 10 min, although at 2, 20 and 30 min it helped on all four), and Filter beat persistence only on Tests A and C (by 2.9% and 1.7%) because Current was itself worse than Persist on three roles. On Test A at 10 min, reporter filtering reduced the error by 9.1% relative to Current, the B3 increment by 4.5% relative to Filter, and input history by 1.0% relative to Filter+B3, giving 0.02098, 8.2% below Persist; these reductions use different baselines and are not additive. On the MSE scale the three Test A reductions are 17.3%, 8.9% and 2.0%; subtracting the white-noise floor, under a tenth of the Persist MSE at 10 min, raises them only to 18.8%, 9.8% and 2.2%. Filter+B3 explains 15.5% of the noise-free Persist error (Fig. 2(b) gives this noise-adjusted skill by horizon, with a 95% cell-bootstrap interval for Filter+B3), and the cell-bootstrap intervals of the three RMSE reductions were 7.4–10.9%, 3.0–6.1% and 0.6–1.4%. The input-history gain was small and not consistent across conditions (two of four on Test A; 0.4% and 0.1% on B and C). The lagged baseline Lag was 2.3% worse than Filter on Test A, but adding the B3 increment to it still reduced the error by 6.9% (four of four concentrations), so on Test A the mechanistic gain holds for both the filter

and the lag representation of the history. In the corrected LOPO (Table III(b), where F+B3 and F+B3+In abbreviate Filter+B3 and F+B3+Input; bold: best per held-out protocol; last column: B3's prior use of that protocol, with $^{\circ}$ marking use at other concentrations), F+B3+Input beat Filter+B3 in all six folds (by 0.1–2.0%) and Lag+B3 beat Lag in all six, but with the 10-min pulse held out, one of the two protocols unseen by B3, Filter+B3 was 1.1% worse than Filter. At longer horizons the mechanistic increment mattered more (Fig. 2(b)): on Test A its skill rose to 21% at 20 and 30 min against 9–13% for Filter, and it anticipated the half-decay crossing of the reporter more often (balanced accuracy 57–58% versus 52–53%; 65–67% versus 54–57% on Test B), although no predictor did so reliably. Explicit input history became harmful on Tests A and B: F+B3+Input fell to –1.7% skill at 20 min on Test A and to –50% and –53% on Test B at 20 and 30 min, although on Test C it lowered the 20- and 30-min error by 3.6% and 3.0%.

B. The Next-Command Window

By phase (Fig. 2), the B3 increment reduced the error of Filter during stimulation (8.9%), early washout (4.6%) and the late phase (6.6%), but raised it in the next-command window from 0.03306 to 0.03544; at 2, 20 and 30 min it lowered the next-command error (by 2.9%, 4.0% and 6.9%), so the failure is specific to the 10-min horizon. That window holds two mixed conditions with ten forecast origins each, five before the 30-min pulse (20 min after the initial 3-min pulse) and five before the 5-min pulse that follows it after 60 min: the increment helped at 0.25 ng/ml (0.03652 to 0.03485) and hurt at 25 ng/ml (0.02920 to 0.03603). Two phase-gated variants fitted on the training protocols did not repair this: a switch to Filter in next-command rows was rejected by the training data, where the increment helped (in-sample MSE 6.3×10^{-4} versus 8.4×10^{-4}), and phase-specific weights (F+B3 (phase) in Fig. 2(b)) raised the next-command error to 0.03571 (held-out mixed fold: 0.03630, versus 0.03537 for Filter+B3 and 0.03302 for Filter). The training protocols contain no probe after a long command, so their next-command windows cannot teach the correction. A frozen descriptive split of the window by the following pulse did not locate the increase. At 25 ng/ml the five origins before the 5-min pulse, 50–58 min after the 30-min pulse, carried 69% of it, below the pre-specified 75%; there the increment raised the error by 35% (0.02737 to 0.03694; worse in 82 of 84 cells): it predicted a rise (mean +0.013) where the reporter fell (–0.009), so its forecasts were too high (mean error +0.021 versus +0.015 for Filter), consistent with B3 overestimating recovery after the long command (Sec. V-E). Before the 30-min pulse it raised the error by 13.5% (0.03092 to 0.03510; mean error –0.003, 95% interval –0.008 to +0.001). The groups also differ in the following pulse, the time since the last pulse and the reporter history, and they are the probe cells of Sec. V-E, so the split is not independent evidence.

TABLE IV
WAITING ON B3 WITH PER-HISTORY CALIBRATION TO 0.94.

Rule	S_3	T_3	S_{30}	T_{30}	ΔT	G (%)	Met
Fixed per history (94/110)	95.0	115.3	95.6	130.4	–	–	20/20
Current obs.	95.0	116.5	95.4	128.2	+0.2	–1	20/20
Smoothed	94.0	108.4	94.6	119.6	–7.7	45	20/20
B3 belief	95.4	109.0	95.0	119.9	–7.2	42	20/20
Oracle	100.0	99.0	98.4	110.4	–	100	–
Noise s.d. 0.01 (same fixed waits)							
Smoothed	94.4	111.8	93.6	122.5	–4.4	26	20/20
B3 belief	95.4	110.5	94.4	121.1	–6.0	35	19/20

C. Waiting under a Per-History Reliability Requirement

Readiness and waiting were simulated on B3 with 25 ng/ml commands and observations every 2 min. Readiness depended almost only on the current reporter, waits of 6–48 min never succeeded, and no state met the criterion without the probe. In Fig. 3(a,b) the smoothed and B3-belief frontiers lie between the fixed waits and the oracle and nearly coincide, whereas the current-observation frontier is no better than the fixed waits after the 3-min command and better only at 90% success or above after the 30-min command. Calibrated to 0.90, every rule, fixed waits included, met 0.9 for both histories in only 7–11 of 20 splits, so a calibration target equal to the requirement leaves no margin. At 0.94 all rules met it in every split at noise 0.005, which is why Table IV tabulates this target, a choice made after the splits were evaluated. The table gives, for each rule (“Current obs.”: the current estimator), the evaluation success S_m (%) and mean completion T_m (min) after the m -min command on the primary split at the reference noise s.d. 0.005 (lower block: 0.01 with the same fixed waits) and, as medians over the 20 splits, the difference $\Delta T = T_{\text{rule}} - T_{\text{fix}}$ in mean completion T over both histories (negative: earlier), the closed share $G = (T_{\text{fix}} - T_{\text{rule}}) / (T_{\text{fix}} - T_{\text{oracle}})$ of the gap to the oracle, and the number of splits (Met) with $S_3, S_{30} \geq 90\%$. Feedback finished 7.2 min (B3 belief) and 7.7 min (smoothed) earlier than per-history fixed waits (94 and 110 min on the primary split), faster in all 20 splits, and closed 42% and 45% of the gap to the oracle. At the other pre-specified targets, 0.90 and 0.92, where the requirement was met in 7–11 and 16–19 of 20 splits, feedback closed 38–42% of the gap (saving 5.5–6.6 min), again faster in all 20 splits. The current-reporter rule closed none of it. Under a pooled requirement (success averaged over both histories), the B3 belief and smoothing closed 33% and 38% of the gap between pooled history-conditioned waits and the oracle (4.8 and 5.5 min; primary split). At noise s.d. 0.01 the B3 belief beat smoothing in all splits at every target (split medians 1.5–1.7 min; 30–35% versus 21–26% of the gap), whereas at 0.005 smoothing was 0.2–0.4 min faster. Larger failure penalties ($T_f = 160$ –260 min, target 0.90) raised the B3 belief's gain over fixed waits from 5.5 to 5.9–8.7 min, with the smoothed and B3-belief rules still ahead of the fixed waits. Sensitivity analyses with pooled calibration (repository) agree: the gain persists at $\kappa = 0.4$ and at a shared calibration margin, at $\kappa = 0.6$

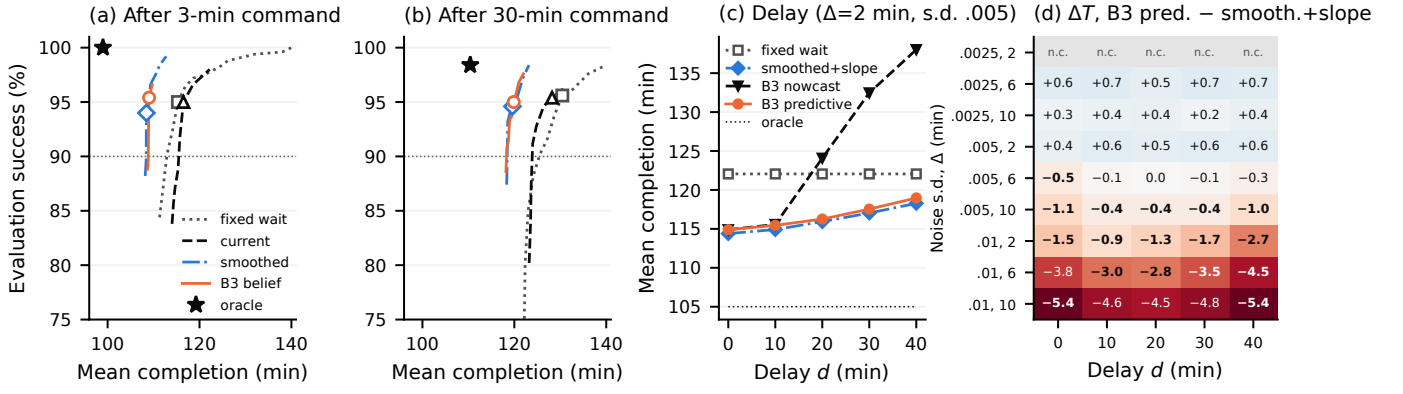


Fig. 3. Waiting on the authors' B3 model (25 ng/ml; task: probe rise $\geq 50\%$ of the naive rise within 20 min). (a,b) Reliability–latency frontiers on evaluation samples (primary split, noise s.d. 0.005; thresholds or waits swept, descriptive), with the operating points calibrated to 0.94 (open markers). (c) Median completion over 20 splits versus feedback delay d (calibration target 0.94 per history). (d) Condition map: median over splits of the paired completion difference, B3-predictive minus smoothed+slope (min; negative: the model is faster) over delay, noise s.d. and sampling interval Δ ; bold: model faster in ≥ 16 of 20 splits at no more than 0.5 points lower success at targets 0.94 and 0.90; n.c.: the B3 rule never reached the target after the 3-min command. One simulated model structure, not measured cells.

even the oracle fails for about 18% of receivers, and under correlated noise the B3 belief needs a likelihood that models the correlation (median success over 20 splits 81% with frozen thresholds and 92% with an AR(1) likelihood, which met 0.9 in 13 of 20 splits), whereas the smoothed rule recovers by recalibration alone (19 of 20).

D. Delay, Noise and Sampling: When the Mechanism Pays Off

With delay, readiness must be predicted. The condition map was first frozen and run with every rule calibrated to 0.90 per history, at which the requirement was met in only part of the splits, and then re-run at 0.92 and 0.94 under a second frozen protocol; we report 0.94, as in Table IV, a target chosen after the first evaluation. The B3 nowcast became slower than the fixed waits from $d = 20$ min on (124–138 versus 122 min), whereas both predictive rules lost only about 4 min between $d = 0$ and 40 min (Fig. 3(c)), so the value of feedback over fixed waits shrank from 7.2–7.9 to 3.1–3.8 min. The condition map (Fig. 3(d)) answers when the model is needed. At noise s.d. 0.01 the B3-predictive rule was faster than smoothing plus slope in every split of every cell, by 0.9–5.4 min, and the smoothing-plus-slope rule became slower than the fixed waits with 10-min sampling, from $d = 20$ min with 6-min sampling and at $d = 40$ min with 2-min sampling (by 0.4–4.7 min), while the B3-predictive rule still beat them (by 1.0–2.8 min). At the reference noise the model helped with 10-min sampling at every delay (0.4–1.1 min) and with 6-min sampling only without delay (0.5 min); with 2-min sampling at the reference noise and with 6- or 10-min sampling at noise 0.0025 the smoothed trend was 0.2–0.7 min faster. At noise 0.0025 with 2-min sampling the B3-predictive rule never reached the target after the 3-min command on calibration (median evaluation success 87.6–90.2%), consistent with weight collapse at low noise, so these five cells are not classified; elsewhere the fixed waits met the requirement in all 20 splits and the B3-predictive and smoothing-plus-slope rules in 19 or 20. The model was

faster in at least 16 of 20 splits in 21 of the 45 cells and also no more than 0.5 points below the smoothing-plus-slope rule's median success in 18 of them; the other three, all at noise 0.01, had 0.6–1.6 points lower median success. Seventeen of the 18 were also selected at 0.90 and are bold in Fig. 3(d); two cells (0.005/6/40 and 0.01/10/30) change class between 0.90 and 0.94. At 0.92, where fixed waits met the requirement in 17 and smoothing plus slope in 13–19 of 20 splits, only 10 of the 17 kept this class (and 0.01/6/0 joined it); in the other seven the B3 rule was still faster in 19 or 20 splits but met the requirement in only 13–15 or had more than 0.5 points lower median success, so the gain at the reference noise depends on the calibration margin. The success condition, on pooled median success, limits but does not rule out trading reliability for speed after one history.

E. Natural Probes in Measured Cells

Two recorded protocols contain later pulses that act as probes at fixed waits (25 ng/ml): in the 3/20 protocol each 3-min pulse comes 20 min after the previous one, with the cell's own first-pulse rise as naive reference (58 cells); in the mixed protocol (84 cells) a 5-min pulse comes 60 min after a 30-min pulse, scored against single 5-min pulse cells (92 cells). Both probes were early by B3's standard: B3 predicted no success for any sample, and every estimator called no cell ready. The measured cells agreed: 19% met the criterion at the first 3/20 probe and 3–9% at later ones, and none at the mixed probe, so the rules made no false ready call but missed the 19%. The rises differed, however: at the mixed probe measured cells reached a median 4% of the naive rise where B3 predicts 31%, consistent with B3 overestimating recovery after a long command. These cells also supply the 25 ng/ml next-command window, in which the increment over-predicted the reporter before this probe (Sec. V-B), so the probe and that forecast error are not independent evidence. After a short pulse B3 erred the other way: at the first 3/20

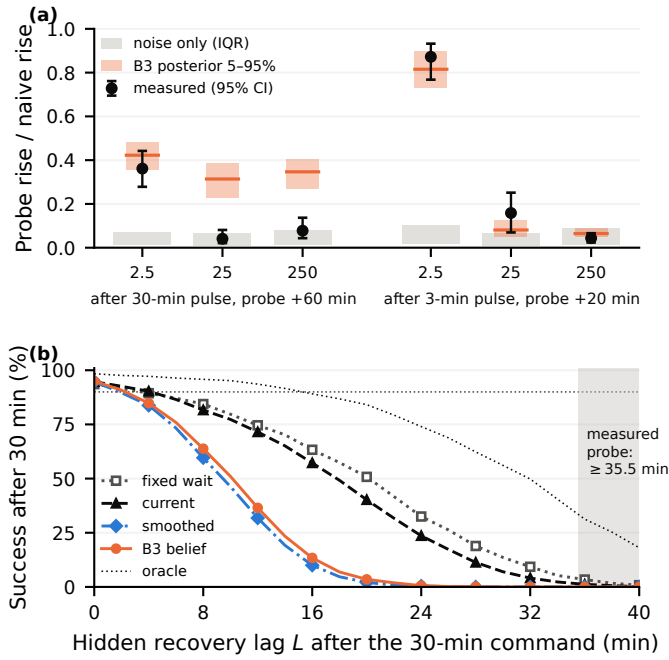


Fig. 4. B3's recovery error on measured cells and its cost for waiting. (a) Probe rise over naive rise at 2.5, 25 and 250 ng/ml after a 30-min pulse (probe 60 min later) and a 3-min pulse (20 min later): measured median with 95% cell-bootstrap interval, B3 posterior predictive 5-95% range (bar; line: median) and the interquartile range expected from noise alone (gray). (b) B3 simulation (target 0.94, noise s.d. 0.005): median success after the 30-min command over 20 splits versus a recovery lag L hidden from the reporter; dotted line: requirement; gray: lags of at least 35.5 min, to which the measured mixed probe corresponds.

probe it underestimated the rise (8% versus 16% of the naive rise). A posterior predictive check across doses (Fig. 4(a)), frozen after an exploratory look at the same quantities, locates the error. Sixty minutes after a 30-min pulse the measured probe rise was 4% and 8% of the naive rise at 25 and 250 ng/ml (95% intervals 2-8% and 4-14%), indistinguishable from no response and below all 1,000 posterior samples (5-95% ranges 23-39% and 27-40%), whereas at 2.5 ng/ml it lay within B3's range (36% versus 36-48%). After a 3-min pulse the measured recovery lay within or at the edge of B3's range at the doses B3 was fitted to (87% versus 73-90% at 2.5 ng/ml, 4% versus 5-9% at 250 ng/ml) and above it only at 25 ng/ml, with overlapping intervals. B3 thus fails where its fit gave it no data, recovery after a long, strong command, which is the regime that matters for reuse, and no reweighting of its samples can repair this. The B3 weights collapsed on measured cells (effective sample size 1.0-1.4), as on the cells B3 was fitted to (1.1-2.3): the posterior's reporter spread (0.001-0.004) lay below the noise floor, whereas the best sample left residuals 3.2-5.3 times the noise floor that were correlated in time (lag-1 autocorrelation 0.22-0.77). A per-cell gain and offset removed the collapse only for the 3/20 cells at 2.5 ng/ml (effective sample size 146); elsewhere residuals stayed 2.5-3.6 times the noise floor, so the pre-specified attribution to deviations that the posterior does not

represent failed for one of four fitted conditions. A real-data filter therefore needs process noise or cell heterogeneity, and the measured noise floor (0.007-0.008) lies between the map's reference and high noise, where with 2-min sampling the map turns from the smoothed trend to the model, which would help only if it were right about recovery (the map was not re-run under the recovery errors of Sec. V-F).

F. When the Model Misjudges Recovery

Every waiting rule above, fixed waits included, is calibrated on B3 outcomes, so none is model-free, and Sec. V-E suggests that B3 recovers too fast after a long command. A frozen simulation therefore delayed the outcomes after the 30-min command by a lag L that the reporter does not show (target 0.94, noise 0.005, 20 splits). Fixed waits and the current-reporter rule kept the per-history requirement up to $L = 2$ min (Fig. 4(b)) (in 20 and 17 of 20 splits), whereas the smoothed and B3-belief rules met it in only 10 and 15 of 20 splits at 2 min, and at $L = 4$ min their success after the 30-min command fell from 95% to 84-85%, against 94% to 89% for the fixed waits, which then met the requirement in 8 of 20 splits. Feedback saves time by probing soon after readiness and so leaves less slack, the time from readiness to the probe, for a recovery error that the reporter hides. The measured mixed probe corresponds to a lag of at least 35.5 min, the most favourable reading of one sub-threshold probe (its 4% rise lies within the interquartile range of rises expected from noise alone); at that lag even the oracle succeeds for at most 31% of receivers after the 30-min command within 120 min, and every rule failed. These are the cells of Sec. V-E and of the next-command window, not independent evidence. Recalibrating only the thresholds on true outcomes restored the requirement up to 6 min for the smoothed and current rules and 4 min for the B3 belief, and recalibrated fixed waits kept it up to 14 min; adding a guard time δ after the threshold crossing (the probe acts at $\min(\tau + \delta, 120)$ with τ from Eq. (6)), calibrated together with the threshold, restored it for all three feedback rules in 18-20 of 20 splits up to $L = 12$ min (recalibration was feasible in every split up to 10 min and in 12 at 12 min), with the smoothed and B3-belief rules 6.2-7.7 min faster than recalibrated fixed waits. This uses the complete outcome curves of the calibration receivers and a common lag, so it bounds what calibration from measured probes can achieve. When the lag also delays the reporter, the smoothed and current rules kept the requirement up to $L = 8$ min (19-20 of 20 splits), the B3 belief up to 4 min and fixed waits not even at 4 min (8 of 20); at noise 0.01 the B3 belief lost more success than smoothing in 17 and 19 of 20 splits at 12 and 16 min (at 16 min the task was feasible in only 4 of 20 splits), while its effective sample size fell to 16 and 6.7. With the model correct, for the slowest quarter of receivers (readiness onset ≥ 98 min after the 30-min command) only the current rule met the requirement in any split (7 of 20), but the B3-belief and smoothed rules raised success after the 30-min command above that of the fixed waits (by 5.5 and 7.1 points, in 18 and 16 of 20 splits; for the slowest half by 2.2

and 0.4 points, in 13 and 10). Feedback thus absorbs recovery variation that the reporter shows, not a recovery error that it hides.

G. Implications for MC Design

A previous command acts on the next one like ISI whose memory sits in the receiver’s biochemistry rather than only in the channel [2], [6]. A fixed wait is a guard interval; conditioning it on the known previous command is the first step (94 and 110 min after the 3- and 30-min commands at target 0.94, primary split), and requiring reliability for each previous command rather than on average lengthens the guard after the long command (106 instead of 102 min at target 0.90, primary split). Feedback then adapts to state variability within each history, much as adaptive detection adapts to residual interference [4], [5], and recovers about 40% of the gap to the oracle at the reference noise. Which estimator feedback needs depends on the noise, the sampling interval and the loop delay: a smoothed reporter with its trend suffices for clean feedback and for frequent feedback at the reference noise, whereas a model is worth its cost for noisy feedback and for sparse feedback at the reference noise, provided it is right about recovery; with a loop delay, either estimate must be predictive. Feedback, however, probes closer to readiness than a guard interval: calibrated on B3, the smoothed and B3-belief rules tolerated no hidden recovery lag, and fixed waits and the current rule only 2 min, far below the lag of at least 35.5 min to which the measured probe corresponds, so a reusable cellular receiver needs a model that is right about recovery or a guard time after the threshold crossing calibrated on probe outcomes (Sec. V-F). The two evidence streams touch but do not combine: on measured cells the mechanistic increment raised the 20–30-min skill to 21% (Fig. 2(b)), and on B3 a loop delay of about 20 min or more required a predictive rule, which a smoothed reporter with its trend supplied unless the noise was high or the sampling sparse. Neither result tests a forecast as a decision rule, and the measured cells suggest that B3 overestimates recovery after a long command (Secs. V-B and V-E, the same cells). A prospective test should randomize 3- versus 30-min preconditioning per chamber, probe at 60–120 min, measure the naive response in same-session controls, calibrate the guard time after the threshold crossing on these probe outcomes and count sessions rather than cells.

H. Evidence Boundaries

Data. The prediction comparisons are post hoc on inspected roles; in the frozen analysis of the main fresh test, history features alone improved on persistence by only 0.7%; Test C shares a session with inspected data, no experimental-day identifiers exist, protocol generalization is therefore not cross-day generalization, and B3 had seen some protocols; the noise floor assumes white reporter noise; the posterior predictive check of B3 recovery was frozen after an exploratory look and uses inspected mixed-protocol cells at 2.5 and 250 ng/ml. *Model.* Waiting results come from one simulated model structure whose posterior samples express parameter uncertainty, not

validated cell heterogeneity; the 20 splits reuse one finite posterior set; the B3 rules share the plant’s structure, which favors them; the recovery errors are stylized time shifts common to all receivers, and the guard-time recalibration assumes complete outcome curves; the targets have no coverage guarantee and a calibration margin is not one. *Task.* $\kappa = 0.5$ was adopted after $\kappa = 0.8$ proved infeasible; the naive response is a simulation counterfactual; amplitude and duration were fixed; the natural probes cover one dose at two fixed waits and score readiness calls, not completion time; the experimental predictors were not tested as decision models, so forecast gains do not imply better waiting decisions.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering plus a fixed calibrated mechanistic increment beat persistence on the validation and all three test protocols, with the increment carrying the gain (it also helped in five of six held-out protocols), whereas filtering alone helped on two of the four and explicit input history added little; the increment raised the pooled 10-min error in the next-command windows of two test conditions, through the 25 ng/ml condition, and a later probe pulse in those cells is consistent with B3 overestimating recovery after a long command, a regime absent from its fit in which the measured recovery lay below every posterior sample at 25 and 250 ng/ml. In B3 simulations with a reliability requirement for each previous command, waiting on a smoothed reporter or the B3 belief closes about 40% of the gap between history-conditioned guard intervals and the oracle at the reference noise; a smoothed reporter with its trend suffices for clean feedback and for frequent feedback at the reference noise, a correct mechanistic model pays off for noisy feedback and for sparse feedback at the reference noise, and with a feedback delay of about 20 min or more, readiness must be predicted rather than nowcast. Feedback saves this time by probing soon after readiness, so in simulation the smoothed and B3-belief rules tolerated no recovery lag hidden from the reporter and fixed waits only 2 min, far below the lag of at least 35.5 min to which the measured probe corresponds; a guard time after the threshold crossing, calibrated on complete outcome curves, restored the requirement for lags up to 12 min while keeping most of the gain, an upper bound, so reusing a cellular receiver needs measured probe outcomes, not only reporter traces, to calibrate such a guard.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, implementation details and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [12], [20], [21]; author files keep their attribution and receive no new license.

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